

1D CNN — Architecture Variant Comparison

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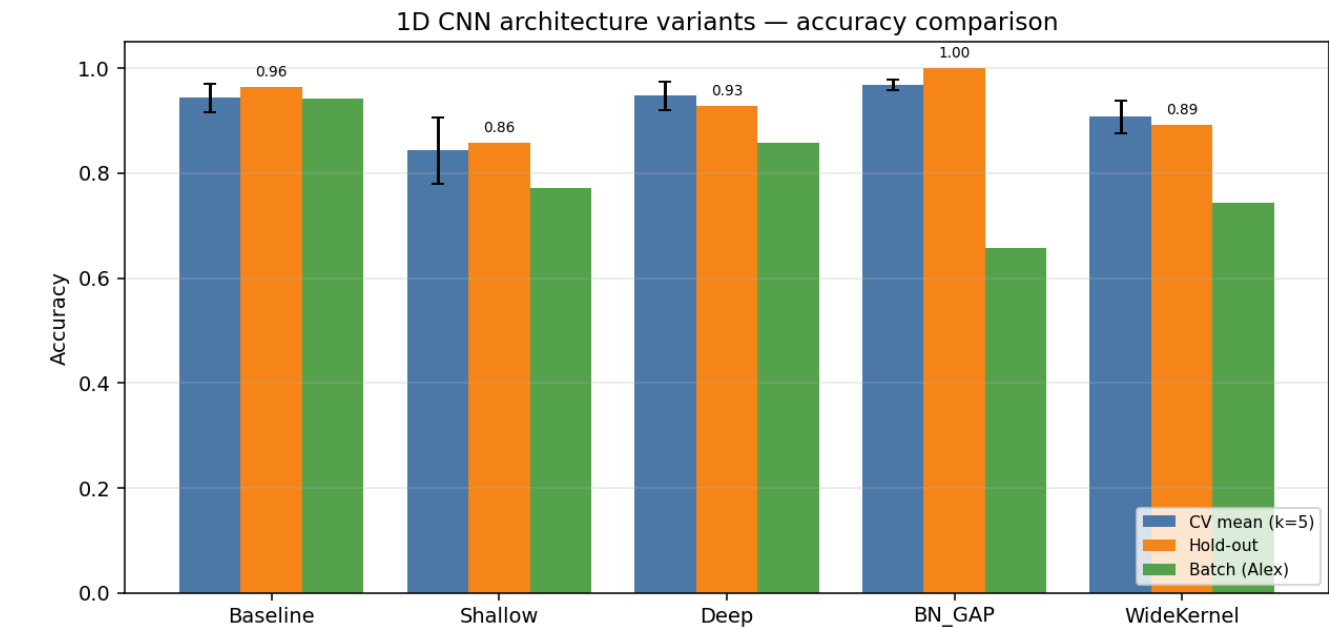
Data & configuration

Training path 1	/home/jestin/ThesisRepo/ML/NewTestData/Alan/Dynamic
Training path 2	/home/jestin/ThesisRepo/ML/NewTestData/Anghad/Dynamic
Training path 3	/home/jestin/ThesisRepo/ML/NewTestData/Harry/Dynamic
Training path 4	/home/jestin/ThesisRepo/ML/NewTestData/Bella/Dynamic
Training path 5	/home/jestin/ThesisRepo/ML/NewTestData/Jestin/Dynamic
Training path 6	/home/jestin/ThesisRepo/ML/NewTestData/Joselyn/Dynamic
Training path 7	/home/jestin/ThesisRepo/ML/NewTestData/Marcus/Dynamic
Training path 8	/home/jestin/ThesisRepo/ML/NewTestData/Raquel/Dynamic
Training path 9	/home/jestin/ThesisRepo/ML/NewTestData/Tash/Dynamic
Batch-test root	/home/jestin/ThesisRepo/ML/NewTestData/Alex/Dynamic
Total trials	276 (train pool 248, hold-out 28)
Classes	7: Double_Lower, Double_Nothing, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll
Sensor channels	176 (sequence length 90)
Resample / filter	90 steps · Butterworth cutoff 10.0 Hz, order 4
Normalisation	minmax
CV folds · shuffle	5 · True
Augmentation	on · copies=5 · amplitude_scale, baseline_offset
Epochs · batch size	15 · 16

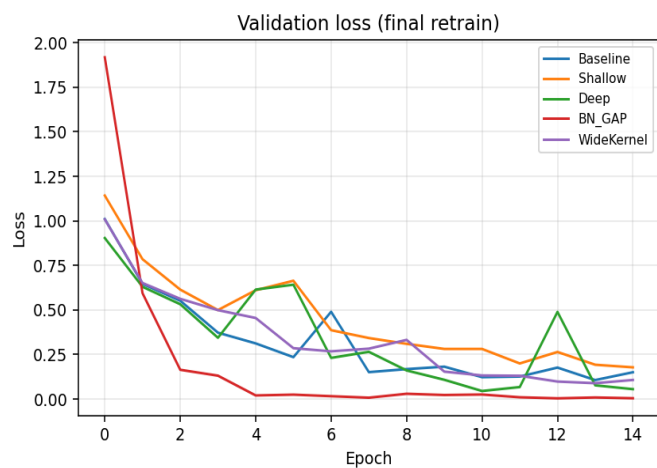
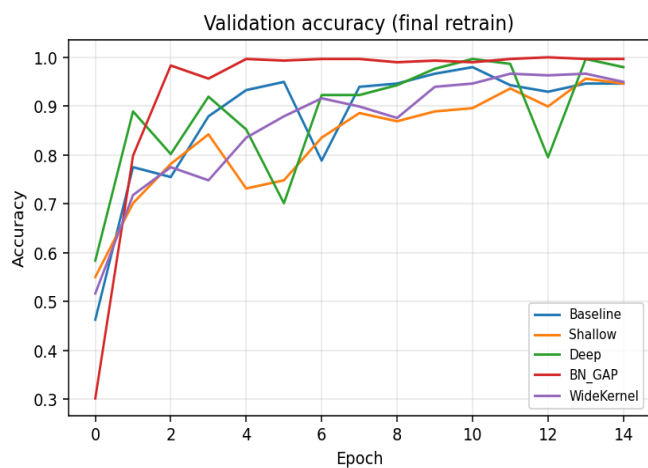
Variant comparison (summary)

Variant	Params	CV mean (k=5)	CV std	Hold-out acc	Hold-out loss	Batch acc	Batch n
Baseline	113,255	0.9433	0.0273	0.9643	0.1233	0.9429	33/35
Shallow	72,487	0.8431	0.0631	0.8571	0.2834	0.7714	27/35
Deep	204,007	0.9476	0.0274	0.9286	0.1763	0.8571	30/35
BN_GAP	57,319	0.9678	0.0098	1.0000	0.0030	0.6571	23/35
WideKernel	131,687	0.9072	0.0303	0.8929	0.3047	0.7429	26/35

Best on hold-out: **BN_GAP** · Best on CV: **BN_GAP**



Training curves (final retrain on full training pool)



Variant: Baseline

Conv1D(32,k=4)→Pool→Conv1D(64,k=4)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	113,255
CV mean ± std	0.9433 ± 0.0273
Hold-out test acc / loss	0.9643 · 0.1233
Batch-test acc	0.9429 (33/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1188	50	0.9600	0.9600	0.9600
2	1188	50	0.9400	0.9800	0.9400
3	1188	50	0.9800	0.9800	0.9800
4	1194	49	0.9388	0.9388	0.9388
5	1194	49	0.8980	0.9796	0.8980

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	4
Double_Nothing	1.000	0.750	0.857	4
Double_Pistol_Recoil	1.000	1.000	1.000	4
Double_Raise	1.000	1.000	1.000	4
Double_Wiggle	0.800	1.000	0.889	4
Double_cmere	1.000	1.000	1.000	5
Drum_Roll	1.000	1.000	1.000	3
macro avg	0.971	0.964	0.964	28
weighted avg	0.971	0.964	0.964	28

Batch test per class (Alex)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	5	5	1.000
Double_Pistol_Recoil	3	5	0.600
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	5	5	1.000
Drum_Roll	5	5	1.000
Overall	33	35	0.943

Variant: Shallow

Single Conv block: Conv1D(32,k=5)→Pool→Flatten→Dense(32)→Dropout(0.3)→Softmax

Trainable params	72,487
CV mean ± std	0.8431 ± 0.0631
Hold-out test acc / loss	0.8571 · 0.2834
Batch-test acc	0.7714 (27/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1188	50	0.8600	0.9000	0.8600
2	1188	50	0.7200	0.7800	0.7200
3	1188	50	0.8600	0.9200	0.8600
4	1194	49	0.8980	0.8980	0.8980
5	1194	49	0.8776	0.9388	0.8776

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	0.667	1.000	0.800	4
Double_Nothing	1.000	0.250	0.400	4
Double_Pistol_Recoil	1.000	1.000	1.000	4
Double_Raise	1.000	1.000	1.000	4
Double_Wiggle	0.750	0.750	0.750	4
Double_cmere	0.833	1.000	0.909	5
Drum_Roll	1.000	1.000	1.000	3
macro avg	0.893	0.857	0.837	28
weighted avg	0.887	0.857	0.834	28

Batch test per class (Alex)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	4	5	0.800
Double_Pistol_Recoil	0	5	0.000
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	5	5	1.000
Drum_Roll	3	5	0.600
Overall	27	35	0.771

Variant: Deep

Three Conv blocks: Conv1D(32,4)→Conv1D(64,4)→Conv1D(128,3)→Flatten→Dense(128)→Dropout(0.4)→Softmax

Trainable params	204,007
CV mean ± std	0.9476 ± 0.0274
Hold-out test acc / loss	0.9286 · 0.1763
Batch-test acc	0.8571 (30/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1188	50	0.9200	0.9200	0.9200
2	1188	50	0.9800	0.9800	0.9800
3	1188	50	0.9400	0.9600	0.9400
4	1194	49	0.9184	0.9184	0.9184
5	1194	49	0.9796	0.9796	0.9796

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	4
Double_Nothing	1.000	0.500	0.667	4
Double_Pistol_Recoil	0.800	1.000	0.889	4
Double_Raise	1.000	1.000	1.000	4
Double_Wiggle	0.800	1.000	0.889	4
Double_cmere	1.000	1.000	1.000	5
Drum_Roll	1.000	1.000	1.000	3
macro avg	0.943	0.929	0.921	28
weighted avg	0.943	0.929	0.921	28

Batch test per class (Alex)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	2	5	0.400
Double_Pistol_Recoil	3	5	0.600
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	5	5	1.000
Drum_Roll	5	5	1.000
Overall	30	35	0.857

Variant: BN_GAP

Conv+BN+ReLU ×3 with GlobalAveragePooling1D head — fewer params, less overfit

Trainable params	57,319
CV mean ± std	0.9678 ± 0.0098
Hold-out test acc / loss	1.0000 · 0.0030
Batch-test acc	0.6571 (23/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1188	50	0.9600	0.9600	0.9600
2	1188	50	0.9800	0.9800	0.9800
3	1188	50	0.9600	0.9800	0.9600
4	1194	49	0.9796	0.9796	0.9796
5	1194	49	0.9592	0.9592	0.9592

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	4
Double_Nothing	1.000	1.000	1.000	4
Double_Pistol_Recoil	1.000	1.000	1.000	4
Double_Raise	1.000	1.000	1.000	4
Double_Wiggle	1.000	1.000	1.000	4
Double_cmere	1.000	1.000	1.000	5
Drum_Roll	1.000	1.000	1.000	3
macro avg	1.000	1.000	1.000	28
weighted avg	1.000	1.000	1.000	28

Batch test per class (Alex)

Class	Correct	Total	Accuracy
Double_Lower	3	5	0.600
Double_Nothing	5	5	1.000
Double_Pistol_Recoil	0	5	0.000
Double_Raise	0	5	0.000
Double_Wiggle	5	5	1.000
Double_cmere	5	5	1.000
Drum_Roll	5	5	1.000
Overall	23	35	0.657

Variant: WideKernel

Conv1D(32,k=8)→Pool→Conv1D(64,k=8)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	131,687
CV mean ± std	0.9072 ± 0.0303
Hold-out test acc / loss	0.8929 · 0.3047
Batch-test acc	0.7429 (26/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1188	50	0.8800	0.9200	0.8800
2	1188	50	0.9600	0.9800	0.9600
3	1188	50	0.9000	0.9200	0.9000
4	1194	49	0.8776	0.8776	0.8776
5	1194	49	0.9184	0.9184	0.9184

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	4
Double_Nothing	1.000	0.500	0.667	4
Double_Pistol_Recoil	1.000	1.000	1.000	4
Double_Raise	0.800	1.000	0.889	4
Double_Wiggle	0.667	1.000	0.800	4
Double_cmere	1.000	0.800	0.889	5
Drum_Roll	1.000	1.000	1.000	3
macro avg	0.924	0.900	0.892	28
weighted avg	0.924	0.893	0.888	28

Batch test per class (Alex)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	5	5	1.000
Double_Pistol_Recoil	0	5	0.000
Double_Raise	5	5	1.000
Double_Wiggle	1	5	0.200
Double_cmere	5	5	1.000
Drum_Roll	5	5	1.000
Overall	26	35	0.743